



IDBI INNOVATE

2026

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IDBI MSME HealthLens

Surface review-worthy thin-file MSMEs without bypassing risk controls.

Team Megalodon · Leader: Bhaskar Itikela

Track 03 — Financial Inclusion / Digital Lending / Credit Decisioning

Consent-backed alternate evidence → Health Card → maker-checker → CAM

Brief about the idea

Balance sheets are thin. Business behaviour is not.

Problem

Thin-file and New-to-Credit / New-to-Bank (NTC/NTB) MSMEs are hard to underwrite on bureau depth alone.

Response

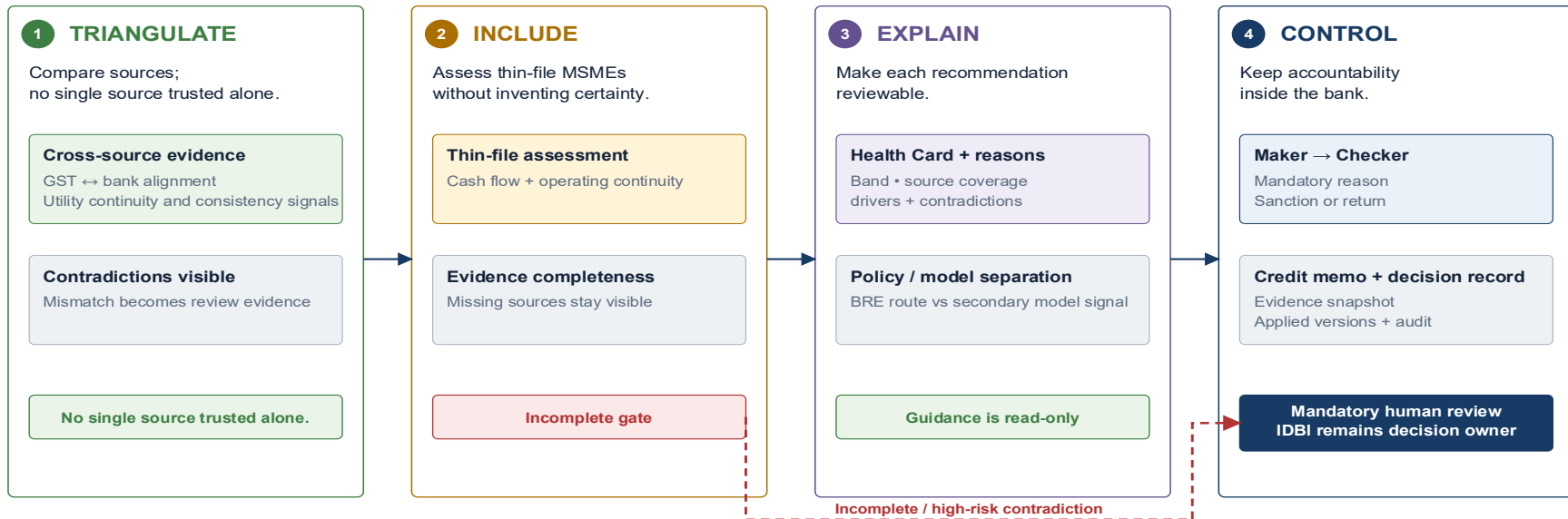
HealthLens turns Account Aggregator (AA), GST, UPI, EPFO, power and fuel evidence into an explainable MSME Financial Health Card.

Control

Recommended route: Approve / Refer / Reject / Incomplete. IDBI remains decision owner. No auto-sanction.

Every recommendation requires maker action and checker sanction. Incomplete, high-risk or low-confidence evidence triggers mandatory manual review. The system does not auto-sanction.

Why HealthLens Is Different



List of features offered by the solution

Assess

Health Card · seven sub-scores · HealthScore · A–D band

Protect

Cross-source anomaly and consistency checks · power/fuel continuity signals · Incomplete gate

Decide

Business Rules Engine (BRE) · limit · product route · maker–checker

Guide

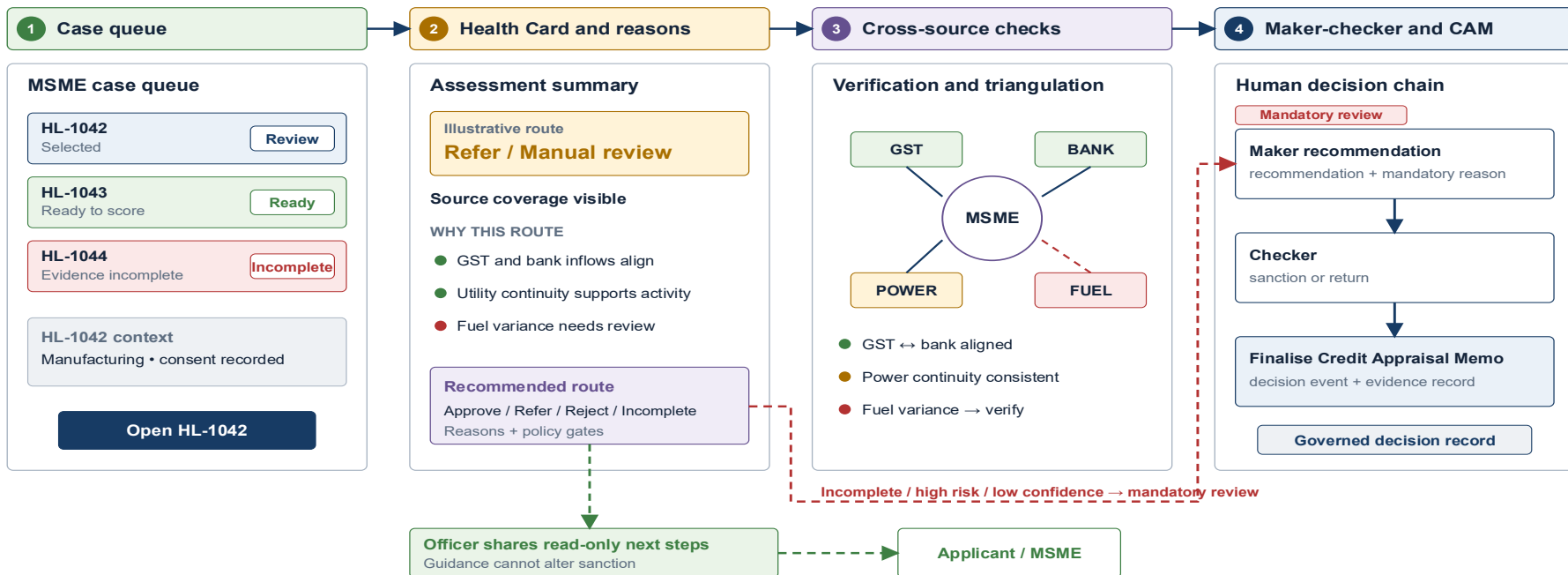
Applicant guidance · path-to-credit · eligibility readiness · officer scenario view

Govern

Credit Appraisal Memo (CAM) · audit · model-card checks · connector status and provenance

Standouts: power/fuel continuity signals and the Incomplete evidence gate keep inclusion controlled.

Officer workflow: evidence to governed decision



Officer workflow in the prototype

1 · Health Card — recommended route

Lakshmi Enterprises Pvt Ltd BAND A Approve Cluster Low High confidence · 100

Private Limited Auto Components Ma... Coimbatore, TN 98m vintage
GSTIN 33VDRID181418... 7/7 sources available

Back to queue

UNDERWRITER VIEW GO Behaviour pattern supports straight-th... RECOMMENDED ₹12L REQUESTED ₹13.5L

HEALTHSCORE **88/100** ML PROXY **90%** GeM Sahay

Health Card Decision Applicant guidance Explainability Fraud & Triangulation Consent & I

WHY THIS CASE IS HERE Go

- GST returns filed on time in ≥ 90% of months
- No material adverse signal detected
- Full alternate-data coverage
- Next action: Export CAM

PRIMARY STRENGTH
GST returns filed on time in ≥ 90% of months

PRIMARY CONCERN
No material adverse signal detected

EVIDENCE
Full alternate-data coverage

NEXT ACTION
Export CAM
Recommendation is ready for officer review and memo export

Sub-scores

Callouts: HealthScore + completeness · recommended route visible · officer owns the decision.

2 · Maker → Checker

Maker — record recommendation

Engine recommends **Approve**. The officer owns the final call; every action is logged to the audit trail.

Approve Refer Reject

Reason for the recommendation (mandatory)...

Maker action (Credit Officer only)

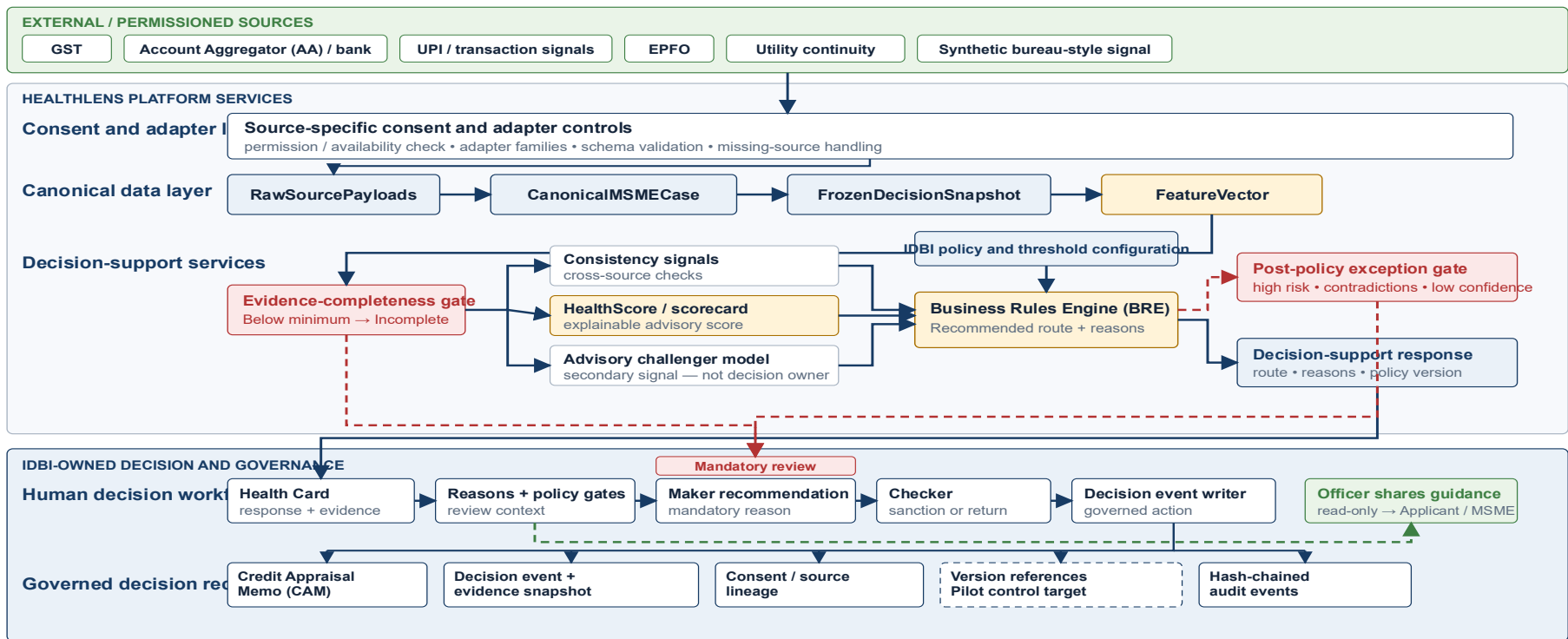
Checker — sanction (four-eyes)

No recommendation recorded yet. A checker can sanction once the maker submits.

Checker note (required to return)...

Callouts: maker records reason · checker sanctions or returns · no auto-sanction.

Architecture diagram of the proposed solution



Technologies to be used in the solution

Decision controls first — UI libraries are implementation detail.

Decision engine

- Versioned TypeScript decision engine
- Canonical Zod contracts
- Deterministic scorecard + Business Rules Engine (BRE)
- FrozenDecisionSnapshot
- Role-based workflow
- Automated tests (116)
- Decision-event recording

Pilot hardening path

- Connector status, source availability and provenance
- Role-Based Access Control (RBAC)
- Audit and observability
- Portable server runtime; pilot deployment subject to IDBI-approved cloud and security controls
- Connector swap surface for sandbox and approved production connectors

Implementation note: TanStack Start, React, Tailwind, shadcn/ui and Recharts support the officer workbench. Connectors are synthetic fixtures today — not live AA / GSTN / DISCOM integrations.

Implementation status and pilot hardening

Effort framing (not a commercial quote) – what is built vs what a sandbox pilot needs.
Status framing only – not a commercial quote or delivery commitment.

BUILT NOW

- Full officer workflow + Health Card
- Power/fuel continuity signals
- Maker-checker + Credit Appraisal Memo (CAM)
- Governance and model-card checks
- Synthetic 600-case book
- Live prototype on Vercel

PILOT HARDENING

- Sandbox and approved production connectors
- Recalibration on bank-labelled outcomes
- Risk-owner policy thresholds
- Officer User Acceptance Testing (UAT)
- Security, consent and observability
- Approved cloud path

Snapshots of the prototype

Explainability + policy gates

Lakshmi Enterprises Pvt Ltd • BAND A • Approve • Cluster Low • High confidence - 100

Private Limited • Auto Components Ma... • Coimbatore, TN • 98m vintage

GSTIN 33VDRID181418... • 7/7 sources available

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HEALTHSCORE 88/100 ML PROXY 90% GeM Sahay

Health Card Decision Applicant guidance **Explainability** Fraud & Triangulation Consent & I

AUDITOR VIEW

Decision is driven by HealthScore + BRE policy gates. The ML proxy is advisory only and does not override IDBI officer judgment.

Policy / eligibility gates

PASS
Bounce history
No hard bounce trigger

PASS
Bureau / DSR
No bureau stress trigger

PASS
GST-bank match
GST and bank credits are within tolerance

PASS
Power / fuel operations
Utility and fuel activity support operations

PASS

Applicant guidance (read-only)

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Health Card Decision **Applicant guidance** Explainability Fraud & Triangulation Consent & I

APPLICANT GUIDANCE - READ-ONLY

What HealthLens would show the MSME — not a sanction screen

Path-to-credit, scheme gaps, and what-if levers only. Go / No-Go and maker-checker remain on the Decision tab for IDBI officers. This is HealthLens coaching language — not a separate borrower product clone.

HEALTHSCORE 88/100 RECOMMENDED LIMIT (INDICATIVE) ₹12L PRODUCT ROUTE GeM Sahay

Authenticity - Adequate

Manufacturing authenticity adequate (power load vs declared turnover).

Scheme readiness Mudra • SVANidhi • CGTMSE • GeM • i-MSME

Mudra

GAPS

Also verify cross-source anomaly and consistency checks in the live prototype. Live: <https://idbi-megalodon.vercel.app>

Open live prototype →

Prototype validation and synthetic benchmark

Synthetic pipeline validation only — not evidence of production predictive performance.

600
MSMEs scored
Deterministic synthetic population

375 / 109 / 108 / 8
Approve · Refer · Reject ·
Incomplete

143 / 143
NTC/NTB kept in funnel
140 advanceable · 3 Incomplete
gate

0.7449
Held-out Area Under the Curve
(AUC)
logistic · synthetic labels

+0.0049
GBM challenger lift
Transparent logistic champion
retained

116
Automated tests
Engine · BRE · analytics · UI

GBM improved synthetic held-out AUC by only 0.0049; the more transparent logistic champion was retained. HealthScore + BRE drive the officer recommendation; the ML proxy is secondary.

Additional Details / Future Development

Future work — same frozen MsmeCase, real/sandbox rails, bank-owned thresholds.

Phased next steps after bank engagement — no firm dates or commercial estimates.

1 · Connectors + policy workshops

Replace synthetic fixtures with sandbox and approved production connectors; agree bank-owned thresholds.

2 · Backtesting + calibration

Recalibrate score bands and BRE thresholds on bank-labelled outcomes.

3 · Officer User Acceptance Testing (UAT)

Tune workflow and thresholds with credit and risk users; keep human-in-loop.

4 · Security, audit, observability

Role-Based Access Control (RBAC), consent artefacts, audit export and monitoring on an approved cloud path.

Out of scope until bank engagement: live production sanctioning, guaranteed AUC, or claiming live AA / ULI / OCEN / GSTN / DISCOM integrations.

Submission links

Live prototype

<https://idbi-megalodon.vercel.app/>

Officer workbench on synthetic data



GitHub (public)

<https://github.com/itikelabhaskar/IDBI-Megalodon>

Source, tests, model card, connector stubs



Demo video (3 minutes)

Google Drive folder (click / scan QR)

Walkthrough: Health Card · checks · Explain · Guidance · maker-checker



Scan QR codes or click the cards. Live · GitHub · Demo Drive.

Prototype scope: synthetic fixtures; no automated sanctioning.

Why shortlist HealthLens

Evaluation evidence for Track 03 — decision support, not auto-sanction.

Track fit

Consent-backed alternate evidence for thin-file MSME credit decisioning with Incomplete as a first-class route.

Differentiation

Power/fuel continuity signals + cross-source consistency checks — no single source trusted alone.

Inclusion control

NTC/NTB stay in the funnel; Incomplete gate refuses forced scores when evidence is thin.

Governance

Every route reaches maker-checker. Low confidence triggers explicit mandatory review. CAM + audit follow checker action.

Execution proof

Live prototype · 600 synthetic MSMEs · 116 tests · frozen case contract · connector status panel.



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Live



GitHub



Demo

Prototype scope: synthetic fixtures · no auto-sanction